Exact values of trigonometric functions of quadrantal angles.

Quadrantal angle	Corresponding point on the Unit circle $P(x,y)=P(\cos\theta,\sin\theta)$	cosine value $\cos\theta$	sine value $\sin\theta$
$\theta=0^\circ$	$(1,0)=(\cos 0^\circ,\sin 0^\circ)$	$\cos 0^\circ=1$	$\sin 0^\circ=0$
$\theta=90^\circ$	$(0,1)=(\cos 90^\circ,\sin 90^\circ)$	$\cos 90^\circ=0$	$\sin 90^\circ=1$
$\theta=180^\circ$	$(-1,0)=(\cos 180^\circ,\sin 180^\circ)$	$\cos 180^\circ=-1$	$\sin 180^\circ=0$
$\theta=270^\circ$	$(0,-1)=(\cos 270^\circ,\sin 270^\circ)$	$\cos 270^\circ=0$	$\sin 270^\circ=-1$
$\theta=360^\circ$	$(1,0)=(\cos 360^\circ,\sin 360^\circ)$	$\cos 360^\circ=1$	$\sin 360^\circ=0$

(i) Observe that x and y coordinates of all points on the unit circle lie between -1 and 1. Hence, $-1\le\cos\theta\le 1$, $-1\le\sin\theta\le 1$, no matter whatever be the value of θ.

(ii) When $\theta=360^\circ$, we have completed one full rotation. Thus, the terminal side coincides with positive x- axis. Hence, sine has equal values at 0° and at 360°. Cosine and other trigonometric functions also follow it.

(iii) If two angles differ by an integral multiple of 360° or 2π, then each trigonometric function will have equal values at both angles.

(iv) Using the values of sine and cosine at quadrantal angles, we have the following generalization geometrically.

At Quadrantal angle	Justification	Generalization
$\sin 0=0$ $\sin\pi=0$	$\sin(0+2n\pi)=0; n\in\mathbb{Z}$ $\sin(\pi+2n\pi)=0; n\in\mathbb{Z}$	$\sin\theta=0\Rightarrow\theta=n\pi; n\in\mathbb{Z}$
$\cos\frac{\pi}{2}=0$ $\cos\frac{3\pi}{2}=0$	$\cos\left(\frac{\pi}{2}+2n\pi\right)=0; n\in\mathbb{Z}$ $\cos\left(\frac{3\pi}{2}+2n\pi\right)=0; n\in\mathbb{Z}$	$\cos\theta=0\Rightarrow\theta=(2n+1)\frac{\pi}{2}; n\in\mathbb{Z}$
$\tan 0=0$ $\tan\pi=0$	$\tan(0+2n\pi)=0; n\in\mathbb{Z}$ $\tan(\pi+2n\pi)=0; n\in\mathbb{Z}$	$\tan\theta=0\Rightarrow\theta=n\pi; n\in\mathbb{Z}$

(v) $\tan\theta$ is not defined when $\cos\theta=0$ and so, $\tan\theta$ is not defined when $\theta=(2n+1)\dfrac{\pi}{2}$, $n\in\mathbb{Z}$.

3.4.2 Trigonometric Functions of real numbers

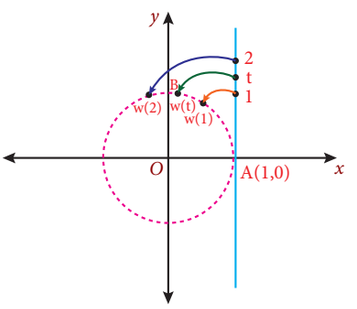


Figure 3.8

For applications of trigonometry to the problems in higher mathematics including calculus and to problems in physics and chemistry, scientists required trigonometric functions of real numbers. This was skillfully done by exhibiting a correspondence between an angle and an arc length denoting a real number on a unit circle.

Consider a unit circle with the centre at the origin. Let the angle zero (in radian measure) be associated with the point $A(1,0)$ on the unit circle. Draw a tangent to the unit circle at the point $A(1,0)$. Let t be a real number such that t is y- coordinate of a point on the tangent line.